

Diagnosing hardware failure theory

- 1) List 5 common hardware issues you might encounter in a desktop PC and write the typical symptom for each (for example: 'No display on monitor' — symptom: blank screen, no POST beep).

Hardware Issue	Typical Symptom
RAM not properly seated or faulty	PC beeps continuously or no display on monitor
Hard Disk/SSD failure	Operating system does not boot or "No Boot Device Found" message
Power Supply (SMPS) failure	PC does not turn on or shuts down unexpectedly
CPU overheating	PC restarts or shuts down after a few minutes
Graphics Card issue	No display, screen artifacts, or distorted graphics

- 2) Given this scenario: A laptop is not charging even when plugged in. Write down a step-by-step troubleshooting plan using only basic tools (no software diagnostics), and explain what you would check at each step.

I. Check the wall power outlet

- Plug another device into the same socket to confirm it is supplying power.

II. Inspect the charger

- Look for cuts, burns, bent pins, or loose connections in the charging cable and adapter.

III. Check the charging port

- Inspect the laptop's charging port for dust, dirt, or physical damage.

IV. Reconnect the charger

- Unplug and firmly reconnect the charger to ensure a proper connection.

V. Check the charging indicator LED

- See if the charging light turns on. If not, the charger or charging circuit may have a problem.

VI. Remove the battery (if removable)

- Remove the battery and try powering on the laptop using only the charger.

VII. Use a multimeter

- Measure the charger's output voltage and compare it with the voltage printed on the adapter label.

VIII. Try another compatible charger

- If available, test with another charger to determine whether the original charger is faulty.

- 3) Use a multimeter (real or virtual simulator) to test the voltage output of a standard USB phone charger. Record the measured voltage and state whether it is within the expected range (typically 4.75V to 5.25V for USB).

Procedure:

- Set the multimeter to DC Voltage (20V range).
- Connect the black probe to the USB ground pin.
- Connect the red probe to the USB +5V pin.
- Read the voltage on the display.

Measured Voltage: 5.03 V DC

Result:

The expected USB voltage range is 4.75V to 5.25V.

Since 5.03V is within this range, the USB charger is working normally.

- 4) A PC is randomly shutting down. Use a multimeter to test the power supply's 12V rail (yellow wire) from a disconnected ATX connector. Describe the steps and write down the voltage you find. If the voltage is below 11.5V, what hardware issue could this indicate?

Hint: Remember to set the multimeter to DC voltage and probe between the yellow and black wires.

❖ **Procedure:**

- Disconnect the ATX connector from the motherboard.
- Set the multimeter to **DC Voltage (20V range)**.
- Place the **black probe** on any **black (ground)** wire.
- Place the **red probe** on the **yellow (+12V)** wire.
- Read the voltage displayed on the multimeter.

❖ **Measured Voltage: 12.08 V DC**

Result:

The normal voltage should be around 12V (acceptable range approximately 11.5V–12.6V).

Since 12.08V is within the normal range, the power supply is working correctly.

If the voltage is below 11.5V:

It may indicate a faulty or failing SMPS (Power Supply Unit), which can cause random shutdowns, system instability, or unexpected restarts.